

Hyokeun Lee

Ph.D.

Assistant Professor

Department of Electrical Engineering and Computer Science

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RESEARCH INTERESTS

- Disaggregated systems, memory security, memory compression, non-volatile memory, computer architecture

EDUCATION

• Seoul National University, Seoul, South Korea

Sep. 2016-Aug. 2021

Ph.D. Electrical and Computer Engineering

- Advisor: Dr. Hyuk-Jae Lee; Co-advisor: Dr. Hyun Kim
- Dissertation: Mitigating Disturbance Errors and Enhancing RMW Performance for PCM

• Seoul National University, Seoul, South Korea

Sep. 2011-Aug. 2016

B.S. Electrical and Computer Engineering

WORK EXPERIENCE

• Department of Electrical Engineering and Computer Science, DGIST

Jan. 2026-Present

Assistant Professor

Daegu, South Korea

• Department of Electrical and Computer Engineering, Ajou University

Mar. 2024-Jan. 2026

Assistant Professor

Gyeonggi, South Korea

• Secure and Advanced Computer Architecture Group, North Carolina State University

Feb. 2023-Jan. 2024

Postdoctoral researcher

Raleigh, NC, USA

- Mentor: Dr. Amro Awad

• Inter-university Research Center, Seoul National University

Sep. 2021-Feb. 2023

Postdoctoral researcher

Seoul, South Korea

- Mentor: Dr. Hyuk-Jae Lee

- I also served the mandatory military service as a Technical Research Personnel.

PUBLICATIONS

#: co-first author; *: corresponding author(s)

• Conference

- Younghoon Ko, Hyemin Park, Hyuk-Jae Lee, **Hyokeun Lee***, "IBEX: Internal Bandwidth-Efficient Compression Architecture for Scalable CXL Memory Expansion," ACM Conference on Supercomputing, July. 2026. [KIISE top conference]
- Jihoon Jang, **Hyokeun Lee***, Hyun Kim*, "EDeN: Enabling Low-Power CNN Inference on Edge Devices Using Prefetcher-Assisted NVM Systems," ACM/IEEE International Symposium on Lower Power Electronics and Design (ISLPED), Aug. 2024. [KIISE top conference]
- Debpratim Adak, **Hyokeun Lee**, Ben Feinberg, Gwendolyn Voskuilen, Clayton Hughes, Huiyang Zhou, Amro Awad, "SEFsim: A Statistically-Guided Fast DRAM Simulator," IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), May. 2024. [KIISE top conference, poster]
- Rahaf Abdullah, **Hyokeun Lee**, Huiyang Zhou, Amro Awad, "Salus: Efficient Security Support for CXL-Expanded GPU Memory," International Symposium on High-Performance Computer Architecture (HPCA), Mar. 2024. [KIISE top conference]
- Faiz Alam#, **Hyokeun Lee**#, Abhishek Bhattacharjee, Amro Awad, "CryptoMMU: Enabling Scalable and Secure Access Control of Third-Party Accelerators," IEEE/ACM International Symposium on Microarchitecture (MICRO), Oct-Nov. 2023 (**Best Paper Candidate**). [KIISE top conference]
- **Hyokeun Lee**, Kwansoek Choi, Hyuk-Jae Lee, Jaewoong Sim, "SDM: Sharing-enabled Disaggregated Memory System with Cache Coherent Compute Express Link," International Conference on Parallel Architectures and Compilation Techniques (PACT), Oct. 2023. [KIISE top conference]
- Shubham Nema, Shiva Kaushik Chunduru, Charan Kodigal, Gwendolyn Voskuilen, Scott Himmert, **Hyokeun Lee**, Amro Awad, Clayton Hughes, "ERAS: A Flexible and Scalable Framework for Seamless Integration of RTL models with Structural Simulation Toolkit," IEEE International Symposium on Workload Characterization (IISWC), Oct. 2023. [KIISE top conference, poster]
- **Hyokeun Lee**, Hyungsuk Kim, Seokbo Shim, Seungyong Lee, Dosun Hong, Hyuk-Jae Lee, Hyun Kim, "PCMCsim: An Accurate Phase-Change Memory Controller Simulator and its Performance Analysis," IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), May. 2022. [KIISE top conference]

- Hyeong Gi Seong, **Hyokeun Lee**, Hyun Kim, Hyuk-Jae Lee, "Analysis of Hardware Prefetchers Suitable for CNN Applications," IEEE/IEIE International Conference on Consumer Electronics-Asia (ICCE-Asia), Nov. 2021.
- **Hyokeun Lee**, Seungyong Lee, Moonsoo Kim, Hyun Kim, Hyuk-Jae Lee, "IMDB: A Low-Cost In-Module Disturbance Barrier for Mitigating Write Disturbance Errors in Phase-Change Memory," Design Automation Conference (DAC, work-in-progress session), July. 2020.
- **Hyokeun Lee**, Donghyeon Lee, Hyuk-Jae Lee, "A Predictive Initialization of Hidden State Parameters in a Hidden Markov Model for Hand Gesture Recognition," IEEE/IEIE International Conference on Consumer Electronics-Asia (ICCE-Asia), June. 2018.

• Journal

- Minseok Seo, Jungi Hyun, Seongho Jeong, Xuan Truong Nguyen, Hyuk-Jae Lee, **Hyokeun Lee***, "OASIS: Outlier-Aware KV Cache Clustering for Scaling LLM Inference in CXL Memory Systems," IEEE Computer Architecture Letters, May, 2025. [SCIE]
- **Hyokeun Lee**, Seungyong Lee, Byeongki Song, Moonsoo Kim, Seokbo Shim, Hyuk-Jae Lee, Hyun Kim, "An In-Module Disturbance Barrier for Mitigating Write Disturbance in Phase-Change Memory," IEEE Transactions on Computers, April. 2023. [SCIE]
- **Hyokeun Lee**, Hyuk-Jae Lee, Hyun Kim, "A Read Disturbance Tolerant Phase Change Memory System for CNN Inference Workloads," Journal of Semiconductor Technology and Science, Aug. 2022. [SCIE]
- Moonsoo Kim, **Hyokeun Lee**, Hyun Kim, Hyuk-Jae Lee, "WL-WD: Wear-Leveling Solution to Mitigate Write Disturbance Errors for Phase-Change Memory," IEEE Access, Feb. 2022. [SCIE]
- Seungyong Lee, **Hyokeun Lee**, Hyuk-Jae Lee, Hyun Kim, "Evaluation of Various Workloads in Filebench Suitable for Phase-change Memory," IEIE Transactions on Smart Processing & Computing, April. 2021.
- **Hyokeun Lee**, Hyunmin Jung, Hyuk-Jae Lee, Hyun Kim, "Bit-width Reduction in Write Counters for Wear Leveling in a Phase-change Memory System," IEIE Transactions on Smart Processing & Computing, Oct. 2020.
- Jinwoo Park, **Hyokeun Lee**, Boyeal Kim, Dong-Goo Kang, Seung Oh Jin, Hyun Kim, Hyuk-Jae Lee, "A Low-Cost and High-Throughput FPGA Implementation of the Retinex Algorithm for Real-Time Video Enhancement," IEEE Transactions on Very Large Scale Integration Systems, Jan. 2020. [SCIE]
- **Hyokeun Lee**, Moonsoo Kim, Hyunchul Kim, Hyun Kim, Hyuk-Jae Lee, "Integration and Boost of a Read-Modify-Write Module in Phase Change Memory System," IEEE Transactions on Computers, Dec. 2019. [SCIE]
- Sunwoong Kim, Hyunmin Jung, Woojae Shin, **Hyokeun Lee**, Hyuk-Jae Lee, "HAD-TWL: Hot Address Detection-based Wear Leveling for Phase-Change Memory Systems with Low Latency," IEEE Computer Architecture Letters, July. 2019. [SCIE]

• Patents

- "Mitigating Write Disturbance Errors of Phase-Change Memory Module," US Patent, No. 11462266, Oct. 2022. (Granted)
- "Semiconductor Memory Device Performing Command Merging and Operating Method Thereof," US Patent, No. 11055025, July. 2021. (Granted)
- "Semiconductor Device for Managing Cold Addresses of Nonvolatile Memory Device," US Patent, No. 10877698, Dec. 2020. (Granted)
- "Semiconductor Device for Managing Wear Leveling Operation of a Nonvolatile Memory Device," US Patent, No. 10713159, July. 2020. (Granted)

• Submitted (current research status)

- Minseok Seo, Seongho Jeong, Hyuk-Jae Lee, **Hyokeun Lee***, IEEE International Symposium on High-Performance Computer Architecture (HPCA), 2026, *under review as double-blind*
- Hyokeun Lee, Rahaf Abdullah, Huiyang Zhou, Abbas Fairouz, Amro Awad, "An Efficient Unified Memory Security Model for CXL Memory-Expanded GPU Systems," IEEE Transactions on Computers, 2025, *under review*
- John McFarland, Hyokeun Lee, Aydin Aysu, Amro Awad, "Secure Integrity for Third-Party IOMMUs," IEEE Transactions on Computers, 2025, *under review*

PROJECT FUNDING GRANTS

• Development of Secure, High-Performance, and High-Connectivity Chiplet-SoC Architecture

Sponsor: MOTIE, South Korea

April. 2025 - Dec. 2027

- Role: PI as lead organization; Funded (own/total): ₩500M / ₩1,000M
- Development of secure chiplet architecture
- Reformulation of LLM data structures and development of supporting architectures for chiplet

• Development of Flash Memory-based AI Processing Unit for On-Device AI

Sponsor: IITP, South Korea

April. 2025 - Dec. 2029

- Role: PI as joint research organization; Funded (own/total): ₩380M / ₩7,220M
- Development of data filtering architecture for efficient LLM near-storage processing

PROFESSIONAL ACTIVITIES

- **IEEE International Conference on Computer Design (ICCD)**
 - PC Member (2025, 2024, 2023, 2022)
- **IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)**
 - Artifact Evaluation Co-Chair (2025)
- **Great Lakes Symposium on VLSI (GLSVLSI)**
 - PC Member (2024)
- **IEEE/IEIE International Conference on Consumer Electronics Asia (ICCE-ASIA)**
 - TPC Member (2025, 2021)
- **IEEE Conference on Artificial Intelligence Circuits and Systems (AICAS)**
 - Reviewer (2023, 2022)
- **International Conference on Electronics, Information and Communication (ICEIC)**
 - TPC Member (2024)
- **ACM Transactions on Architecture and Code Optimization**
 - Reviewer (2025, 2024)
- **IEEE Transactions on Computers**
 - Reviewer (2025, 2024)
- **IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems**
 - Reviewer (2025)
- **IEEE Transactions on Dependable and Secure Computing**
 - Reviewer (2025)
- **IEEE Computer Architecture Letters**
 - Reviewer (2025)
- **Springer The Journal of Supercomputing**
 - Reviewer (2025, 2024)
- **IEIE Transactions on Smart Processing and Computing**
 - Reviewer (2024, 2023, 2021)
- **Elsevier Microelectronics Journal**
 - Reviewer (2022)
- **Elsevier Materials Research Bulletin**
 - Reviewer (2021)

AWARDS/MENTIONS

- Best Paper Runner-Up, MICRO 2023 (Mentioned).
- First Prize in Haedong Best Paper Award (Academic Research Work), South Korea, Nov. 2021.

INVITED TALKS

- "*Towards Large-Scale Computing Systems via Resource Disaggregation*," Seoul National University of Science and Technology, South Korea, Dec. 2024
- "*Disaggregation and its Security for Large-Scale Computing Systems*," Korea University, South Korea, Aug. 2023.
- "*CXL-enabled Sharing in a Multi-Host Disaggregated Memory System*," Seoul National University GoGE Workshop: Future Generation Security Computing Systems, Aug. 2023.
- "*Introduction to Computer Architecture Simulators and Use of Gem5*," North Carolina State University (ECE 096), United States, Mar. 2023.
- "*Bitwidth Reduction of Write Counters of Wear Leveling in a Phase-Change Memory System*," KSPC, South Korea, Sep. 2022.
- "*Introduction to Computer Architecture Simulators and Examples*," University of Seoul, South Korea, Jun. 2022.

TEACHING LECTURES

- **Advanced Computer Architecture**

- Graduate class, teaching advanced topics of microarchitectues like out-of-order processors
- Teaching years: 2025

- **VLSI System Design**

- Forth year undergraduate class, teaching lower-level issues in digital systems
- Teaching years: 2024, 2025

- **Computer Organization and Architecture / Computer Architecture**

- Third year undergraduate class, teaching basics of microarchitecture and memory subsystems
- Teaching years: 2024, 2025, 2026

- **Digital System Design**

- Third year undergraduate class, teaching digital system design methods using Verilog
- Teaching years: 2024, 2025

TECHNICAL SKILLS

Programming: C++, C, Verilog, Python

Architecture Simulators: NVMain, PCMCsim, DRAMsim3, McSimA+, MacSim, gem5, SST

Hardware Interface (Transaction Level): AXI-Lite/-Full/-Stream, CXL, PCIe, I2C, JEDEC DDRx

Languages: English (Professional working proficiency), Korean (Native), Chinese (Bilingual)